

Objective - To determine the value of acceleration due to gravity and radius of gyration using bar pendulum.

Apparatus required - Bar pendulum, Stopwatch and meter scale.

Formula used -

A. The general formula of the time period for bar pendulum is given by following equation.

$$T = 2\pi \sqrt{\frac{\frac{k^2 + l^2}{L}}{g}} = 2\pi \sqrt{\frac{L_2 + L_1}{g}}$$

$$g = \frac{4\pi^2(L_1 + L_2)}{T^2} = \frac{4\pi^2 L}{T^2} \quad \text{--- (1)}$$

where, L_1 : distance between C.G. and suspension point,
 L_2 : distance between suspension and oscillation point.

$$L = L_1 + L_2 = L + \frac{k^2}{L}$$

g : acceleration due to gravity, T : time period.

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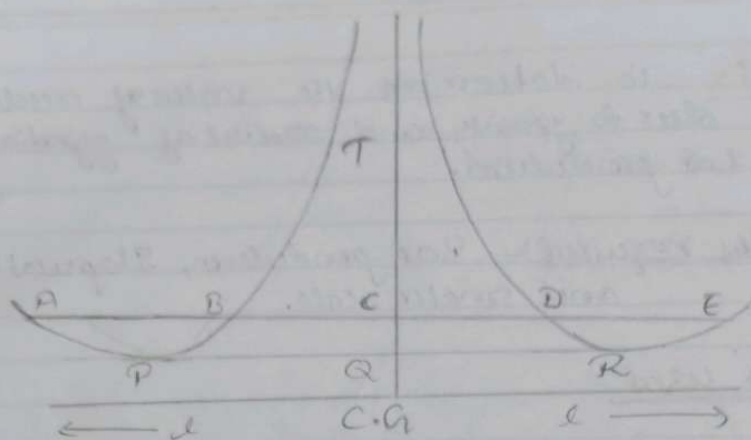


Fig. 2

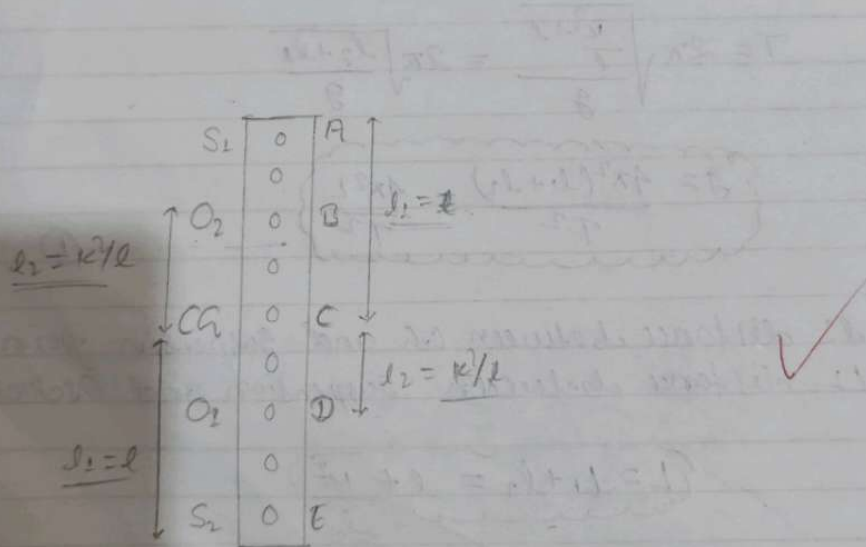


Fig. (1)

B. The time period is minimum when $l = \pm k$, in this situation the equation (1) becomes as:

$$T_{\min} = 2\pi \sqrt{\frac{2k}{g}}$$

$$g = \frac{8\pi^2 k}{T_{\min}^2}$$

(2)

or

where, k : radius of gyration, $T_{\min}$: minimum time period.

The value of 'g' can be calculated using equations (1) & (2).

The values of L , T , k and $T_{\min}$ are obtained using graph between T and L for bar pendulum which is shown in following figure.

From Figure (1) and (2)

(a) $L_1 = AC + CD$, $L_2 = EC + EB$ and $L = (L_1 + L_2)/2$, T = time at

(b) $k = (PC + QR)/2$ and $T_{\min}$ = time at Q

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Calculation from graph,

$$L = (AD + EB) / 2 = \frac{62 + 63}{2} = 62.5 \text{ cm} \quad T = \dots \text{ sec},$$

$$K = PR / 2 = \frac{60}{2} = 30 \text{ cm} \quad T_{\min} = \dots \text{ sec},$$

$$L_1 = (AC + CE) / 2 = \dots 45$$

$$L_2 = (BC + CD) / 2 = \dots 17.5$$

$$1. \quad g_1 = \frac{4\pi^2 L}{T^2}$$

$$2. \quad g_2 = \frac{8\pi^2 K}{T_{\min}^2}$$

$$3. \quad g = \frac{g_1 + g_2}{2}$$

$$(1) \quad g_1 = \frac{4\pi^2 L}{T^2} = \frac{4 \times 3.14 \times 62.5 \times 10^{-2}}{(1.00)^2} = 8.94 \text{ m/sec}^2$$

$$(2) \quad g_2 = \frac{8\pi^2 K}{T_{\min}^2} = \frac{8 \times 3.14 \times 30 \times 10^{-2}}{(1.55)^2} = 9.9 \text{ m/sec}^2$$

$$(3) \quad g = \frac{g_1 + g_2}{2} = \frac{8.94 + 9.9}{2} = \frac{18.84}{2} = 9.42 \text{ m/sec}^2$$

$$(4) \quad K = \sqrt{L_1 L_2} = \sqrt{45 \times 17.5} = \sqrt{787.5} = 28.06 \approx 28 \text{ cm}$$

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C The radius of gyration can be obtained with following formula.

$$K = \sqrt{I_1 I_2}$$

$$\text{where, } I_1 = (AC + CE)/2, I_2 = (BC + CD)/2$$

Observations:-

1. Least count of the stop watch = _____ sec
2. Least count of the meter scale = _____ cm

3. Table for d and T

S.No.	d (cm)	t (time taken for 20 oscillations)	$T = t/20$
For First half side of the bar			6.
1.	45	38.5	1.925
2.	40	38.5	1.925
3.	35	31.5	1.575
4.	30	31	1.55
5.	25	31	1.55
6.	20	32	1.6
7.	15	34	1.7
8.	10	39	1.95
9.	5	53	2.625
For Second half side of the bar			
10.	-5	55	2.75
11.	-10	39.5	1.975

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12.	-15	34.4	1.72
13.	-20	33	1.635
14.	-25	31.5	1.56
15.	-30	30.5	1.55
16.	-35	31.5	1.575
17.	-40	32.5	1.61
18.	-45	33	1.65

Results:The acceleration due to gravity (g) = $\approx 9.7 \text{ m/sec}^2$ Radius of gyration (k) = 30 cm (from calculation)Precautions:-

(1) The motion of the pendulum should be in a vertical plane. while taking the time, start taking observations after two oscillations to avoid any irregularity of motion.

2. The amplitude of oscillation should be small.

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